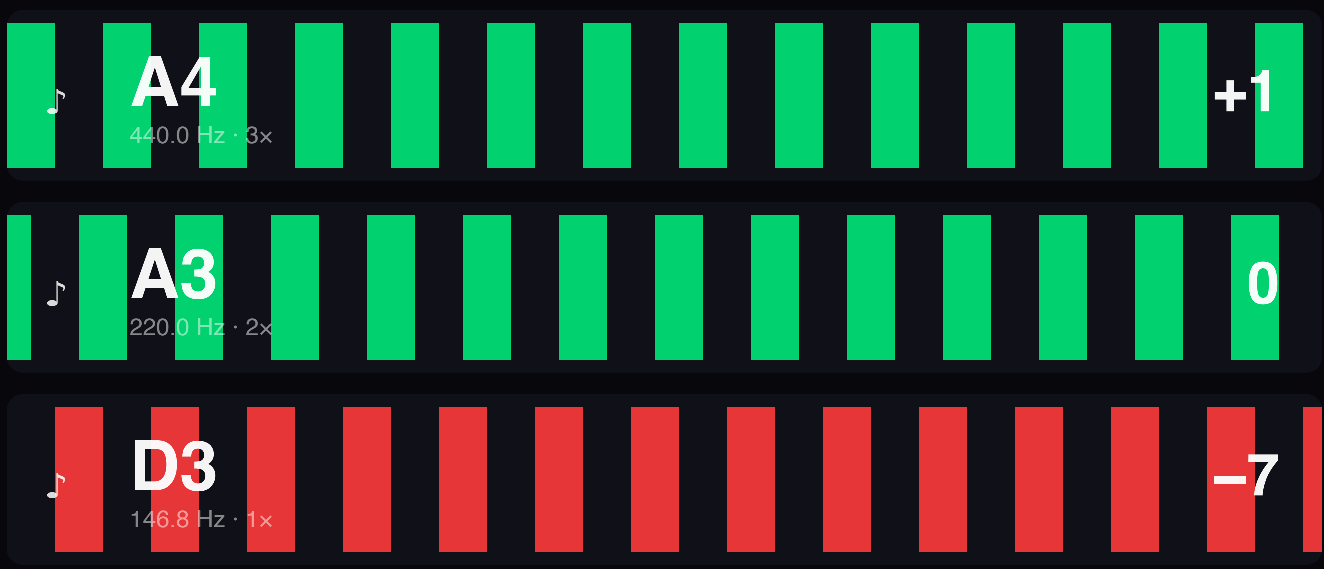
V-Tune

User Guide

A precision strobe tuner for handpans
and other multi-modal instruments.



The three-band strobe display — still + green when locked, drifting + red when out of tune.

v-tune-handpan.vercel.app · v1.0.1

FFT peak detection + phase-rate Goertzel analysis

Contents

1. Welcome

What V-Tune is and why it exists

V-Tune is a precision strobe tuner built for handpans and other multi-modal instruments — drums, gongs, bells, anything where a single strike excites several pitches at once. Most chromatic tuners give up on these instruments because they're built around one fundamental. V-Tune is built around three: a fundamental, an octave, and an octave-plus-fifth (the 12th), all rendered as independent strobe bands so you can read every partial without them fighting each other.

Under the hood, V-Tune uses FFT peak detection plus phase-rate Goertzel analysis — the FFT tells you which frequencies are loud, the Goertzel measures how stable each one is, frame to frame. You see the result as classic strobe-pattern bars: motion = drift, still = locked.

This guide walks through every part of the app. It's short — V-Tune is a small tool, and the parts that exist exist for a reason. Read it once, then keep tuning.

2. Quick start

From cold app to tuned note in about a minute

1. Grant microphone access

On first launch V-Tune asks for the mic. Allow it. You can change device later in Settings Input (open Settings with the gear in the teal icon bar).

2. Pick a note

The menu loads open. In the Tuning / Scale controls pick a scale from the SCALE dropdown and tap a note, or stay on Chromatic and use OCT – / OCT + to set the octave. The three foundation strobe bands (1×, 2×, 3×) update instantly.

3. Press Let's Go

The green button lives at the bottom of the menu (or the bottom of the mobile slide-up). Audio starts. Strike your instrument. The bars freeze when you're in tune; they drift left when you're flat, right when you're sharp.

4. Read the cents number on the right of each band

It reads 0 (and the band washes green) when you're in tune, within ±5¢ by default. The signed number is exactly how many cents off you are.

Sample strobe band — in tune

3. The menu & utility bar

Where every control lives — and how it gets out of your way

All of V-Tune's controls live in one place: the menu. It looks different on a wide screen versus a phone, but the contents are the same — a teal utility bar up top, the tuning / scale controls, and Let's Go pinned to the bottom.

On desktop & landscape tablet

A full-height panel slides out from the right edge. Open and close it with the burger button (three lines in a rounded square) at the top-right of the header. The menu always loads open on launch so everything is to hand.

To give the strobe more room while you tune, the menu auto-hides after 20 seconds of no interaction with it. Any tap, drag or scroll inside the menu resets the timer. If you want it to stay put, hit the pin — that disables auto-hide until you unpin.

If the menu hides while a stopwatch session is running, a compact stopwatch readout appears in the header next to the burger, so your timing stays visible. Tap it to reopen the menu.

On phone & portrait tablet

There's no burger or side panel. Instead the controls live in a bottom quick-pick panel that slides up. Collapsed, it's a single bar with a soft purple glow showing the currently-selected note (e.g. "D3"), centred. Tap it to slide the panel up — this pushes the canvas up, which shrinks responsively to make room. It also auto-hides after 20 seconds of no interaction, and has its own pin to keep it open. Full details in section 7.

The teal utility bar

A slim teal-tinted row of icon buttons sits at the top of the menu (and inside the mobile slide-up). The left group, in order:

- Settings — opens the Settings modal (section 9).
- Stopwatch — reveals or hides the stopwatch (section 10).
- Spectrum Analyser (equaliser-bars icon) — reveals or hides the Spectrum Analyser (section 8).

On the right sits the light / dark theme toggle (sun / moon). On mobile the pin icon also lives on the right, just left of the theme toggle. Any active toggle — stopwatch on, spectrum on, pinned — gets a teal highlight so you can see its state at a glance.

4. The strobe display

Where you actually read the tune

The strobe display is the centre of V-Tune. Each horizontal stripe is one band — a frequency the app is tracking. By default you get three foundation bands stacked top-to-bottom:

The display is theme-aware. At rest — when the mic is quiet — the background is a light grey in light mode, or a soft charcoal (not pure black) in dark mode. The moment the mic picks up signal, the background darkens toward a deep near-black for maximum bar contrast, then eases back to its resting tone when you stop. The neutral text (note names, cents) flips light or dark to stay legible against whichever background is showing.

- 3× — the 12th (octave + fifth). Top of the stack.
- 2× — the octave.
- 1× — the fundamental. Bottom of the stack.

A thicker cyan separator marks the boundary between foundation bands and any custom bands you add (via the Spectrum Analyser).

Reading a band

Each band shows you five things at a glance:

- ♪ icon (far left) — the per-band pitch pipe. See section 5.
- Note label — e.g. “D3”. The note this band is tuned to.
- Frequency — the exact target in Hz, under the note label.
- Strobe bars — the moving red/green pattern. Motion direction tells you sharp vs flat.
- Cents readout (far right) — the deviation in cents, signed. Reads 0 (and the band washes green) when in tune.

Colours, and what they mean

- In tune (within ±5¢)
- Selected band
- ♪ Pipe — beep mode / detected note
- Out of tune
- ♪ Pipe — continuous tone
- Scale "ding" highlight

Motion = drift

Bars moving left your input is flat (pitch too low). Tune up. Bars moving right your input is sharp (pitch too high). Tune down. Bars holding still you’re locked. When a band is in tune it gets a dark-green wash and its cents readout settles on 0.

The bars are green in tune and red out of tune, and their blur tracks how far off you are: crisp and sharp when locked, blurring more the further out of tune you drift (and during the unstable attack transient of a fresh strike). Settings Blur sets the ceiling on that softness.

Peak hold + decay

V-Tune holds the strobe pattern for ~4 seconds after each strike, then fades it out over a second. This means you can let the note die away and still read the tune from the held pattern — no need to keep re-striking. A louder hit resets the hold timer.

5. The pitch pipe (♪)

A reference tone for every band, three clicks deep

Each strobe band has a ♪ icon on its left edge. Click it to hear the band's exact target frequency. Click again to cycle modes:

♪	Off	Silent. The default state.
♪	Tone	Continuous sine reference. Plays the band's frequency forever (until you click again). Useful for hum-along tuning.
♪	Beep	Silent until the band detects a strike — then it fires a brief reference beep at the target frequency. So as you tune each strike triggers a comparison tone you can match against. This is the LinoTune-style mode.

Clicking the ♪ icon on a different band moves the pipe to that band — same mode, new frequency. Only one band can be piping at a time. To stop completely, click the active band's ♪ until it cycles back to Off.

6. The tuning & scale controls

Where you tell V-Tune which note to listen for

On desktop the tuning and scale controls sit directly in the menu, below the teal bar. On phones and portrait tablets they split: the scale picker, note grid and Let's Go live in the slide-up quick-pick panel (section 7), while Reference A4 and Tolerance move into Settings Tuning. Either way the picker has two modes:

AUTO

Tap AUTO to let V-Tune auto-detect whichever note you play, rather than pinning it to a note you've chosen. Handy when you're sweeping across an instrument and don't want to keep re-selecting.

Chromatic mode

A piano-keyboard layout — 7 naturals + 5 sharps. Tap any key to tune to that note in the current octave. Use the OCT – / OCT + buttons below to drop or raise the octave. This is the default mode on launch.

Scale mode

Pick a pre-saved handpan scale (Kurd, Amara, Celtic, etc.) from the SCALE dropdown above the keyboard. The note picker switches from a chromatic keyboard to just the notes in that scale, each with a fixed octave matching the physical instrument. The ding (root) is highlighted in purple. Switching scales auto-selects the ding.

Note naming: in scale mode the accidentals follow the scale's convention (Kurd uses flats, Amara uses sharps). In chromatic mode they follow your Settings Notation preference.

Live note indicator

While audio is running, the note V-Tune is currently picking up from the microphone gets a soft cyan glow ring around its button. This is purely informational — it doesn't change which note you're tuning against. Useful for quickly orienting yourself when you strike an unfamiliar partial.

Tuning reference — PURE vs EQUAL

Above the note layout is a PURE / EQUAL toggle that decides what the three foundation bands are measured against:

- PURE (default) references each band against an exact integer multiple of the fundamental — 1×, 2×, 3×. A handpan whose partials are tuned to those pure ratios reads 0 on every band. This is the acoustically correct reference for handpan partial tuning.
- EQUAL references each band against the nearest equal-tempered note instead. On a pure handpan the compound-fifth (3×) band then reads about +2 cents — the real, constant difference between a pure 3:1 fifth and a tempered fifth. Use this if you tune your partials to equal temperament.

The octave (2×) band reads the same either way — octaves are an exact 2:1 in both systems. Only the compound fifth differs.

Reference A4, Tolerance & Let's Go

Reference A4 sets concert pitch (default 440 Hz) and Tolerance sets how close counts as in tune (default ±5 cents). On desktop these sit in the menu alongside the picker; on mobile they move into Settings Tuning. Let's Go — which starts and stops audio — is pinned to the bottom of the menu on desktop, and the bottom of the slide-up on mobile.

7. Mobile quick-pick panel

The controls, condensed for thumbs

On phones and portrait tablets there's no burger or side panel. The controls live in a bottom quick-pick panel that slides up. Collapsed, it's a single bar with a soft purple glow showing the currently-selected note (e.g. "D3"), centred. Tap the bar to slide the panel up — this pushes the canvas up, which shrinks responsively to make room.

Opened, it carries the same teal utility bar (settings, stopwatch, spectrum, pin, theme) and the scale / note controls:

- SCALE dropdown — same scales as desktop, above the note grid.
- Chromatic / Scale note grid — a responsive grid of notes, each labelled with its octave; OCT –/+ in chromatic mode.
- Let's Go — pinned to the bottom of the slide-up.

Like the desktop menu, the slide-up auto-hides after 20 seconds of no interaction so the strobe gets full width, and it has its own pin (in the teal bar) to keep it open. When it re-collapses it just shows the selected note, centred, ready to tap again. Reference A4 and Tolerance live in Settings Tuning on mobile.

8. Spectrum Analyser & Isolation windows

See the full frequency content, then isolate the bits you care about

Toggle the Spectrum Analyser with its icon (equaliser bars) in the teal utility bar. It appears under the strobes as a frequency-domain view across the audible range — a real-time picture of every harmonic your instrument is producing, with two isolation strobe bands beneath it for fine-tuning partials.

Turning the Spectrum Analyser on always restores the two default isolation windows — the teal and purple ones — every time, even if you previously cleared them. So it's never revealed empty.

Isolation windows (ISO)

V-Tune starts with two isolation windows ready to go — a teal one and a purple one. Each is a bracket on the spectrum; V-Tune finds the loudest peak inside the bracket and drives a dedicated strobe band from it, shown beneath the spectrum. The band carries the same colour as its bracket, so it's always clear which window feeds which band.

Drag either end of a bracket to move it; the frequency / note / cents readout follows so you can line an edge up against a note. This is how you tune partials that aren't the fundamental, octave or 12th — say, a particular overtone. Remove a window with its × button, and draw a new one any time with Shift + drag (or touch-hold then drag on mobile) across the spectrum; you can have up to two at once, splitting the band area 50/50. A re-added window reclaims the freed colour slot.

9. Settings

Every knob, what it does

Settings is a modal, opened by the gear in the teal utility bar. It's organised into labelled teal-header sections, each laid out in two columns — the setting's name and a short description on the left, its control on the right.

Input

Microphone	Which audio input device V-Tune listens to. Default uses the system mic.
Microphone Sensitivity	Input gain in dB (MIC +/–). Bump it up if your input is quiet, drop it if you're clipping.
Hum	Mains-hum notch filter: Off / 50 Hz (UK/EU) / 60 Hz (US). Notches out mains hum plus its harmonics.

Tuning (mobile only)

On phones and portrait tablets, Reference A4 and Tolerance live here. On desktop these controls sit directly in the menu instead, so this section only appears on mobile.

Reference A4	Concert pitch. Default 440 Hz. Set to 442 for some orchestral work, 432 if you're into that.

Tolerance (±cents)	How close you have to be before V-Tune calls you in tune. Default is ±5 cents.

Strobe Preferences

Brightness	How vivid the red/green strobe bars are. Lower for ambient lighting, higher for stage / sunlight.
Blur	Edge softness of the bars — sharp when locked, automatically softer when way out of tune. This sets the ceiling on that softness.
Speed	How fast the strobe reacts: 0.5× / 1× / 2× / 3× / 5×. Higher = livelier, lower = calmer / easier to read.

Accessibility Options

High contrast	Boosts contrast throughout the UI for easier reading.
Larger text	Scales up the interface text.
Show tour again	Re-runs the interactive onboarding tour from the start (section 11).

10. Stopwatch

Time your tuning sessions

Toggle the stopwatch on and off with the icon in the teal utility bar. When on, its panel appears pinned at the bottom of the menu, just above Let’s Go (on mobile, just above the slide-up). Start, stop and reset controls sit on the panel. It counts continuously even if you toggle the panel off and back on.

On desktop, if the menu auto-hides while a session is running, a compact stopwatch readout appears in the header next to the burger so your timing stays visible. Tap it to reopen the menu.

11. Theme, notation, and the onboarding tour

A few preferences worth knowing about

Light / dark theme

Flip between light and dark with the sun / moon icon on the right of the teal utility bar. Fresh installs default to light mode; switch to dark whenever you prefer. The strobe display follows the theme too — see section 4.

Notation

Pick how accidentals are labelled in Settings Notation: ♯ sharps, ♭ flats, Do solfège, or DE German naming (which uses H for B and B for B♭).

The onboarding tour

V-Tune ships with an interactive, learn-by-doing tour. It spotlights each part of the UI and waits for you to actually perform the action before moving on — open the menu, open Settings and walk its sections, reveal the stopwatch and the Spectrum Analyser, pin the menu, pick a scale, then Let's Go. It runs automatically on first launch.

Want it again? Open Settings and hit Show tour again under Accessibility Options — it restarts the whole guided tour from the top, any time.

Staying up to date (desktop)

The macOS, Windows and Linux apps update themselves. When a newer version is published, V-Tune notices on launch and offers a one-click Install & Restart — no need to redownload or reinstall. (iOS and Android update through their stores; the web app refreshes itself.)

12. Tips & troubleshooting

Things to try if something feels off

"The bars are jittery / never lock."

Background noise is usually the culprit. Open Settings with the gear, then (a) lower Microphone Sensitivity until just your strikes register, and (b) set Hum to your local mains frequency (50 Hz UK/EU, 60 Hz US).

"The cents number is right but the strobe disagrees."

They're measuring different things. The cents number is a median-filtered, EMA-smoothed reading designed to be a stable digit you can read. The strobe is raw phase rate — it shows instantaneous motion. Trust the strobe for fine adjustments; trust the cents number for the overall verdict.

"It's telling me I'm in tune but I'm clearly not."

Check Reference A4. If it's set to something exotic (442, 432) you'll be tuning against a different concert pitch — it's in the menu on desktop, or Settings Tuning on mobile. Also check you're on the right note: handpans have rich overtones, and it's easy to accidentally lock onto a partial that isn't the fundamental. The live note indicator on the note grid (cyan glow) helps catch this.

"The strobe is too lively / too sluggish."

Settings Speed is your friend. Drop it for a calmer reading, raise it for a more responsive one.

"I want to tune a partial that isn't 1×, 2×, or 3×."

Turn on the Spectrum Analyser, find the peak you care about, and shift+drag (or touch-hold drag on mobile) a bracket around it. V-Tune will generate a dedicated strobe band for that partial.

"My phone screen has a notch / home indicator."

Handled — V-Tune respects iOS safe-area insets on the header, the quick-pick slide-up, and the menu. The strobe never hides behind the notch or the home-indicator strip.

That’s the whole app. Strike a note, watch it lock in.